

SESA 6067

Spacecraft Orbital Mechanics

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1. Chapter 1: Orbital Maneuvers

1.1. Maneuvers

A maneuver is defined as **A controlled change in the velocity (magnitude and/or direction) of a spacecraft to alter its orbit**. Each mission contains multiple maneuvers and those maneuvers can be made of the following building blocks:

- Change orbit shape.
- Change orbit orientation (plane).
- Change of orbit phasing (anomaly).

There can be multiple reasons for maneuvers within a mission, including orbital transfers, maintenance and disposal. The goal of mission design is to **minimize ΔV within mission constraints**, these constraints being:

- Transfer time.
- Forbidden regions (e.g. radiation belts).
- Pointing requirements (panels, instruments).

1.2. Types of Maneuvers

This module will cover various types of coplanar transfers and plane change maneuvers, these are:

- Coplanar transfers
 - (Bi-elliptic) Hohmann transfer
 - Phasing maneuvers
 - Non-Hohmann transfers
 - Apse line rotation
- Plane change maneuvers
 - Inclination change
 - RAAN change

1.3. Impulsive Maneuver Assumption

For this module, each maneuver is assumed to be impulsive (kick maneuver). This assumes that the burn time is negligible in comparison to the orbital time. It is important to note that **ΔV cost of a mission is the sum of all maneuvers**. Further note that for this topic **ΔV is a vector and so must be added correctly**.

1.4. Hohmann Transfers

A Hohmann transfer is a form of co-planar maneuver that can be used to raise or lower an orbit. These types of maneuvers are typically used during launch to increase the height of an orbit to the desired altitude.

1.4.1. Circular Hohmann Transfer

A circular Hohmann transfer is used to move between two circular orbits via one intermediate elliptical orbit. The derivation for the orbital velocities of each orbit can be done via the angular momenta of each orbit. To start with a diagram of a circular Hohmann transfer is shown in **Figure 1**.

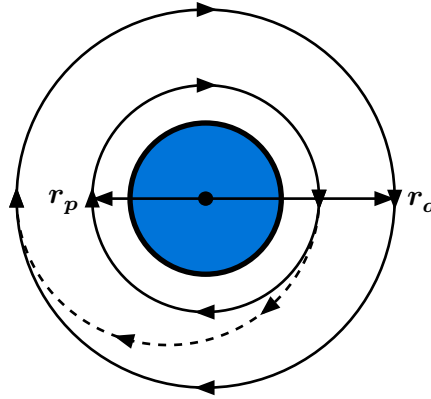


Figure 1: Schematic diagram of a circular Hohmann Transfer.

In order to calculate the ΔV required for the two burns in this maneuver the polar orbit equation can be used, shown in **Eq. 1**, note that at perigee for the elliptical transfer orbit $\theta = 0^\circ$

$$r = \frac{p}{1 + e \cos(\theta)} = \frac{a(1 - e^2)}{1 + e \cos(\theta)} = \frac{h^2/\mu}{1 + e \cos(\theta)} \rightarrow r_p = \frac{h^2/\mu}{1 + e} \quad (1)$$

The expression in **Eq. 1** can be combined with the eccentricity equation shown in **Eq. 2**.

$$e = \frac{r_a - r_p}{r_a + r_p} \quad (2)$$

Combining both expressions **Eq. 1** and **Eq. 2** yield the following expression shown in **Eq. 3**.

$$r_p = \frac{a(1 - e^2)}{1 + e} \quad e = \frac{r_a - r_p}{r_a + r_p} \rightarrow h = \sqrt{2\mu \frac{r_a r_p}{r_a + r_p}} \quad (3)$$

Where h is the specific angular momentum ($m^2 s^{-1}$) of the orbit and r_p and r_a are the radius at perigee and apogee shown in **Figure 1**. Now consider the general expression for specific angular momentum shown in **Eq. 4**.

$$h = \vec{r} \times \vec{v} = rv \sin(\phi) \quad (4)$$

Where $\vec{r}$ and $\vec{v}$ are the position and velocity vectors and the angle ϕ is the angle between these two vectors. Importantly, $\phi = 0^\circ$ at the apogee and perigee points in the elliptical

cal transfer orbit. Combining **Eq. 3** and **Eq. 4** the velocities at r_p and r_a can be written in the form shown in **Eq. 5**.

$$v_a = \frac{h}{r_a} \quad v_p = \frac{h}{r_p} \quad \text{Where } h = \sqrt{2\mu \frac{r_a r_p}{r_a + r_p}} \quad (5)$$

Note that if $r_p = r_a$, meaning the transfer orbit is circular, then the expression simplifies to the following known expressions for circular orbits, shown in **Eq. 6**.

$$\text{If } r_a = r_p = r \quad \text{Then} \quad h = \sqrt{\mu r} \quad v = \sqrt{\frac{\mu}{r}} \quad (6)$$

1.4.2. Elliptical Hohmann Transfer

The derivation on the previous page is **not specific to circular orbits** as all of the equations used are general orbital equations. For this reason the equations can also be used for **transferring between two elliptical orbits** given they are coaxial (the Apse line of both orbits is the same).

However, there is a caveat in that it is not initially clear which transfer orbit is the most efficient. This is exacerbated if the two orbits are not in the same direction. This scenario would require calculating the deltaV of all 4 possible orbits (2 transfer orbits in each direction).

The two possible orbits are shown in **Figure 2**.

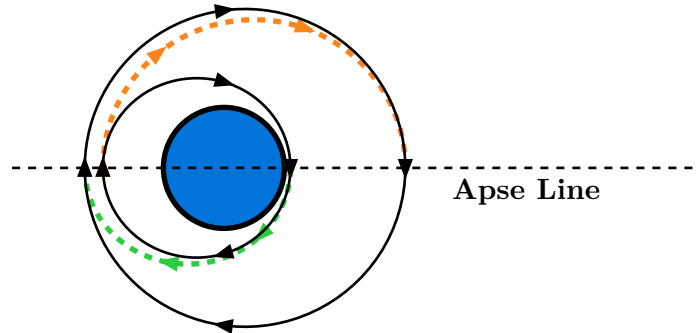


Figure 2: Schematic diagram of elliptical Hohmann transfers.

1.4.3. Parabolic and Hyperbolic Hohmann Transfer

The same equations can also be used for transferring between an elliptical/circular orbit to a hyperbolic or parabolic orbit. Note that both of the orbits must lie on the same apse line as one another, examples of these types of orbit are shown in **Figure 3**.

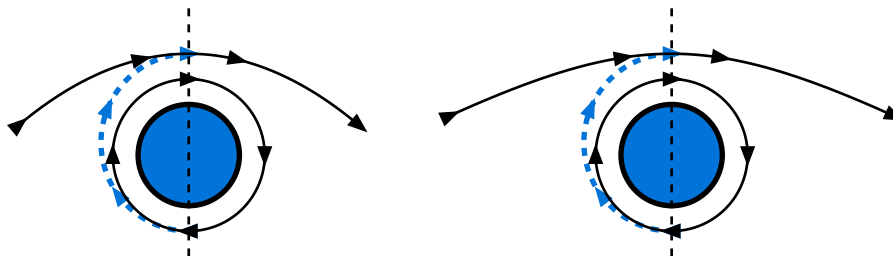


Figure 3: Schematic diagram of parabolic Hohmann transfer [Left] and hyperbolic Hohmann transfer [Right].

1.4.4. Bi-elliptic Hohmann Transfer

If the maximum number of burns is increased to three, then in some cases the **bi-elliptic Hohmann transfer** may be more efficient than a basic Hohmann transfer. This type of maneuver is shown in **Figure 4**

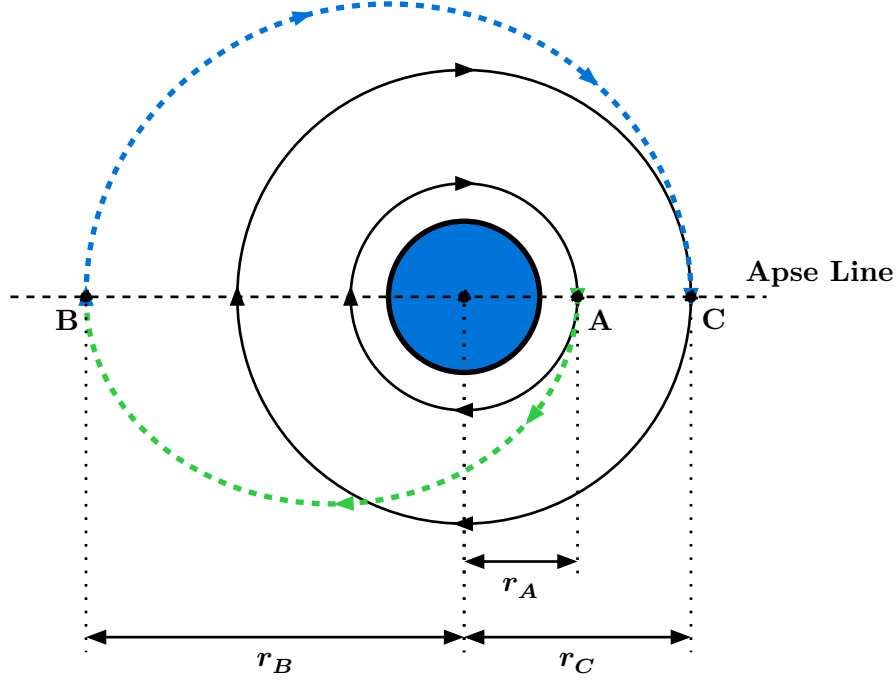


Figure 4: Schematic diagram of bi-elliptic Hohmann transfer.

As r_b goes to infinity, the ΔV required at **B** goes to zero, allowing for a cheaper burn. However, a bi-elliptic Hohmann transfer is not always more efficient and so a derivation is required.

To start with, it is observed there are two transfer orbits T_1 from $A \rightarrow B$ and T_2 from $B \rightarrow C$, the perigee and apogee radii of these two transfer orbits is shown in **Eq. 7**.

$$T_1 \begin{cases} r_a = r_B \\ r_p = r_A \end{cases} \quad T_2 \begin{cases} r_a = r_C \\ r_p = r_B \end{cases} \quad (7)$$

From these definitions the change in velocity at point **A** (ΔV_A) can be calculated and is shown in **Eq. 8** (Note, V_0 = velocity in origin orbit, V_1 = perigee velocity in T_1).

$$V_0 = \sqrt{\frac{\mu}{r_A}} \quad V_1 = \frac{h}{r_p} = \frac{1}{r_p} \sqrt{2\mu \frac{r_a r_p}{r_a + r_p}} = \sqrt{2\frac{\mu}{r_p^2} \frac{r_a r_p}{r_a + r_p}} = \sqrt{2\frac{\mu}{r_A} \frac{r_B}{r_A + r_B}} \quad (8.1)$$

$$\Delta V_A = V_1 - V_0 = \sqrt{2\frac{\mu}{r_A} \frac{r_B}{r_A + r_B}} - \sqrt{\frac{\mu}{r_A}} = \sqrt{\frac{\mu}{r_A}} \left[\sqrt{2\frac{r_B}{r_A + r_B}} - 1 \right] \quad (8.2)$$

$$\therefore \Delta V_A = V_0 \left[\sqrt{2\frac{r_B}{r_A + r_B}} - 1 \right] \quad \text{Where } V_0 = \sqrt{\frac{\mu}{r_A}} \quad (8.3)$$

The derivation for the change in velocity at point **B** (ΔV_B) can be determined similarly, and is shown in **Eq. 9** (Note, V_1 = apogee velocity in T_1 , V_2 = apogee velocity in T_2).

$$V_1 = \frac{h}{r_a} = \sqrt{2\frac{\mu}{r_B} \frac{r_A}{r_A + r_B}} \quad V_2 = \frac{h}{r_a} = \sqrt{2\frac{\mu}{r_B} \frac{r_C}{r_C + r_B}} \quad (9.1)$$

$$\Delta V_B = V_2 - V_1 = \sqrt{2\frac{\mu}{r_B} \frac{r_C}{r_C + r_B}} - \sqrt{2\frac{\mu}{r_B} \frac{r_A}{r_A + r_B}} \quad (9.2)$$

$$\therefore \Delta V_B = \sqrt{2\frac{\mu}{r_B}} \left[\sqrt{\frac{r_C}{r_C + r_B}} - \sqrt{\frac{r_A}{r_A + r_B}} \right] \quad (9.3)$$

The derivation for the change in velocity at point **C** (ΔV_C) can be determined similarly, and is shown in **Eq. 10** (Note, V_2 = perigee velocity in T_2 , V_3 = velocity in destination orbit (further note that $V_C = V_2 - V_3$ to keep ΔV +ve)).

$$V_2 = \frac{h}{r_p} = \sqrt{2\frac{\mu}{r_C} \frac{r_B}{r_C + r_B}} \quad V_3 = \sqrt{2\frac{\mu}{r_C}} \quad (10.1)$$

$$\Delta V_C = V_2 - V_3 = \sqrt{\frac{\mu}{r_C}} \left[\sqrt{2\frac{r_B}{r_C + r_B}} - 1 \right] \quad (10.2)$$

The total ΔV of this maneuver would be the sum of **Eq. 8**, **Eq. 9** & **Eq. 10**, however some substitutions are made to make the resulting equations simpler, these substitutions are shown in **Eq. 11**.

$$\text{Let } \beta = \frac{r_B}{r_A} \quad \alpha = \frac{r_C}{r_A} \quad V_0 = \sqrt{\frac{\mu}{r_A}} \quad \therefore \quad (11.1)$$

$$\Delta V_A = V_0 \left[\sqrt{2\frac{\beta}{\beta + 1}} - 1 \right] \quad (11.2)$$

$$\Delta V_B = V_0 \sqrt{\frac{2}{\beta}} \left[\sqrt{\frac{\alpha}{\beta + \alpha}} - \sqrt{\frac{1}{\beta + 1}} \right] \quad (11.3)$$

$$\Delta V_C = V_0 \sqrt{\frac{1}{\alpha}} \left[\sqrt{2\frac{\beta}{\beta + \alpha}} - 1 \right] \quad (11.4)$$

Note that in **Eq. 11** a common term of V_0 can be extracted. Adding these expressions whilst multiplying V_B by $\sqrt{\frac{2}{\beta}}$ and V_C by $\sqrt{\frac{1}{\alpha}}$ yields **Eq. 12**.

$$\Delta V = \Delta V_A + \Delta V_B + \Delta V_C \quad (12.1)$$

$$= V_0 \left[\sqrt{\frac{2\beta}{\beta + 1}} - \underset{\substack{\uparrow \text{Multiply by } \beta/\beta}}{\mathbf{1}}} + \sqrt{\frac{2\alpha}{\beta(\beta + \alpha)}} - \underset{\substack{\uparrow \text{Multiply by } \alpha/\alpha}}{\mathbf{1}}} + \sqrt{\frac{2}{\beta(\beta + 1)}} + \sqrt{\frac{2\beta}{\alpha(\beta + \alpha)}} - \sqrt{\frac{1}{\alpha}} \right] \quad (12.2)$$

↑ Combine together

$$\therefore \sqrt{\frac{2\alpha^2}{\alpha\beta(\beta + \alpha)}} + \sqrt{\frac{2\beta^2}{\alpha\beta(\beta + \alpha)}} = \sqrt{\frac{2(\alpha + \beta)}{\alpha\beta}} \quad -1 - \sqrt{\frac{1}{\alpha}} = -\frac{1 + \sqrt{\alpha}}{\sqrt{\alpha}} \quad (12.3)$$

$$\therefore \sqrt{\frac{2\beta^2}{\beta(\beta+1)}} - \sqrt{\frac{2}{\beta(\beta+1)}} = \sqrt{\frac{2}{\beta(\beta+1)}}(\beta-1) \quad (12.4)$$

Combining together the simplified expressions in **Eq. 12** yield the final expression for the ΔV cost for a bi-elliptic Hohmann transfer shown in **Eq. 13**

$$\Delta V = V_0 \left[\sqrt{\frac{2(\alpha + \beta)}{\alpha\beta}} - \frac{1 + \sqrt{\alpha}}{\sqrt{\alpha}} + \sqrt{\frac{2}{\beta(\beta+1)}}(\beta-1) \right] \quad (13.1)$$

$$\text{Where :} \quad V_0 = \sqrt{\frac{\mu}{r_A}} \quad \beta = \frac{r_B}{r_A} \quad \alpha = \frac{r_C}{r_A} \quad (13.2)$$

By plotting β against α , it is possible to see when the bi-elliptic Hohmann transfer is more efficient than a normal Hohmann transfer, this plot is shown in **Figure 5**.

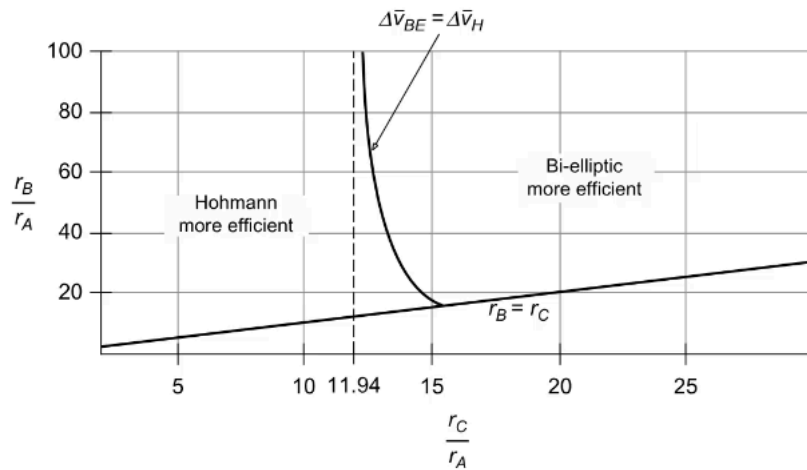


Figure 5: Plot of β against α for a bi-elliptic Hohmann transfer.

In **Figure 5** the straight line labelled $r_B = r_C$ is a normal Hohmann transfer. Specifically if $\alpha > 11.94$ then a **bi-elliptic Hohmann transfer is more efficient than a normal Hohmann transfer**.

